Modeling Inhomogeneous Broadening through Spin-Spin Interactions in Matrix Isolated Spin Ensembles

Isaac Cross
Northwestern University, Evanston, IL

Introduction

A large field of physics involves looking at an atom's internal "spin states" and the transitions between these states. These spin states act as incredibly sensitive probes and can be used to search for new physics. The sensitivity to many of these searches is proportional to $\frac{1}{T_2^*\sqrt{N}}$. We can improve N, number of spins probed, by isolating a huge number of them in a matrix. However, current matrix isolation experiments have a very small "coherence time", T_2^* due to inhomogeneous broadening. The source of this broadening is debated, some claim it's from crystal defects, while others claim its from spins interacting. The goal of my project, therefore, is to model these spin interactions and the resulting inhomogeneous broadening using Monte Carlo Methods.

Spin-Spin Interaction Model

- Most experiments in atomic physics involve probing well defined transitions between spin states.
- This project models atomic transitions between two spin states labeled $|F, m_F\rangle$.
- One drives this transition with a resonant frequency ν_0.



- Ideally: all atoms in matrix have same energy splitting $h\nu_0$. This would result in zero inhomogeneous broadening.
- However, neighboring spins can interact and shift each other's energy. This interaction is largely through magnetic interactions.
- Model spins as dipole magnets (like a bar magnet)

Magnetic field at $\boldsymbol{r}$ relative to magnetic dipole:

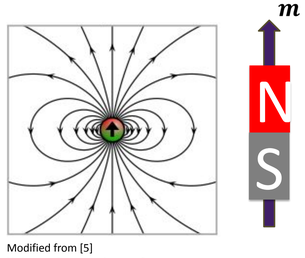
$$\boldsymbol{B}(\boldsymbol{r},\boldsymbol{m}) = \frac{\mu_0}{4\pi r^3}[3\hat{\boldsymbol{r}}\cdot(\boldsymbol{m}\cdot\hat{\boldsymbol{r}}) - \boldsymbol{m}]$$

Dipole "magnetic moment":

$$\boldsymbol{m} = g_F m_F \mu_B \hat{\boldsymbol{m}}$$

Hyperfine Landé g-factor:

$$g_F \approx g_J \frac{F(F+1)-I(I+1)+J(J+1)}{2F(F+1)}, \quad g_J \approx 2.0023$$

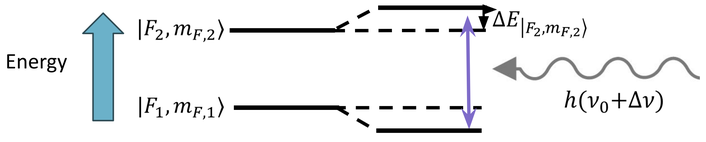


Modified from [5]

For the sake of this project, I, J, F, and m_F can all be thought of as input parameters defined by the atom and transition we are modeling. Look at [4] if you want a physical understanding.

- Spin states experience "Zeeman shifts" in magnetic fields

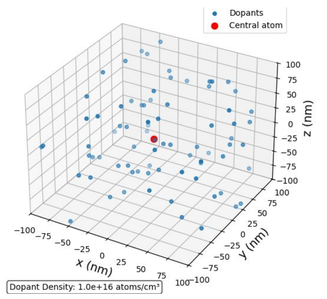
$$\text{Zeeman Shift: } \Delta E_{|F,m_F\rangle} = \mu_B g_F m_F B_z$$



$$\Delta\nu = \frac{\Delta E_{|F_2,m_{F,2}\rangle} - \Delta E_{|F_1,m_{F,1}\rangle}}{h} = \gamma_{\text{eff}} B_z, \quad \gamma_{\text{eff}} = \frac{\mu_B}{h}(g_{F_2} m_{F,2} - g_{F_1} m_{F,1})$$

Broadening Model

- Inhomogeneous broadening can be thought of as the distribution of $\Delta\nu$ in an ensemble of atoms (dopants).
- To estimate this distribution, we use a Monte Carlo method
- We start by defining a test cell with side length L. We place a test atom at the origin and distribute $N_{neighbors}$ dopants, located at $\boldsymbol{r}_i$, in the cell corresponding to the atom density ρ.



$$N_{neighbors} \sim \text{Poisson}(\lambda) - 1, \quad \lambda = \rho L^3$$

$$\boldsymbol{r}_i = (x_i, y_i, z_i) \sim \text{Uniform}\left(\left[-\frac{L}{2}, \frac{L}{2}\right]^3\right), \quad i = 1,2,\ldots,N_{neighbors}$$

Test Cell Spin States Model:

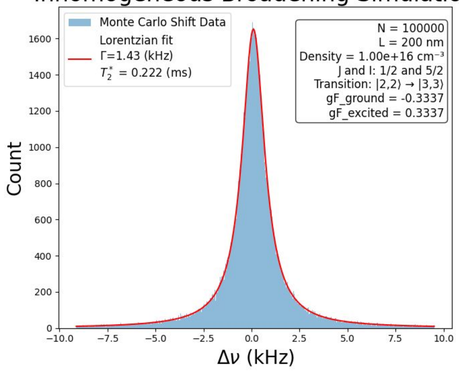
- We choose an atomic transition to model at the central test atom
$$|F_1, m_{F,1}\rangle \rightarrow |F_2, m_{F,2}\rangle$$
- Surrounding atoms are prepared into $|F_1, m_{F,1}\rangle$ using two "pumping efficiencies"
  - F efficiency p_F : fraction atoms forced into F_1; remaining assigned randomly to F_1 or F_2
  - m_F efficiency p_{m_F} : fraction atoms forced into $m_{F,1}$; remaining assigned randomly over $m_F = -F_1, \ldots, +F_1$

Spin Magnetization Model:

- Experiments typically apply a static magnetic field, $\boldsymbol{B_0}$, along the z-axes, which aligns the magnetic moments of our spins along z.
- To model variation in this alignment, we add alignment efficiency p_{align}: fraction of spins with magnetization set along z; remaining assigned random orientation on unit sphere.

Determining Broadening:

- Using the distribution of spins in our test cell we determine the total magnetic field at the origin and corresponding frequency shift:
$$B_z(0,0,0) = \hat{\boldsymbol{z}} \cdot \sum_{i=1}^{N_{neighbors}} \boldsymbol{B}(-\boldsymbol{r}_i, \boldsymbol{m}_\mathbf{i}) \qquad \Delta\nu = \gamma_{\text{eff}} B_z(0,0,0)$$
- By generating multiple test cells and collecting $\Delta\nu$ each time, we can create a distribution of how often certain $\Delta\nu$'s occur. Extracting inhomogeneous broadening



Fit with Lorentzian to extract inhomogeneous broadening linewidth Γ:

$$\text{L}(\Delta\nu) = \text{A}\,\frac{\Gamma/2}{(\Delta\nu)^2 + (\Gamma/2)^2}$$

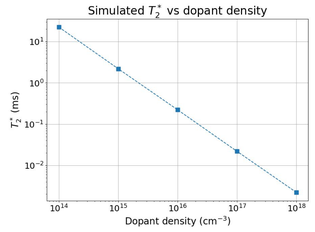
Coherence time:

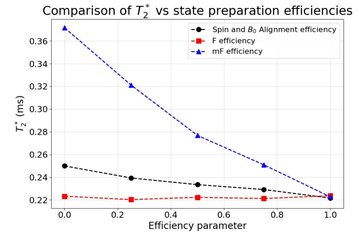
$$\text{T}_2^* \equiv \frac{1}{\pi\Gamma}$$

Results

How Broadening Model Scales With Key Inputs:

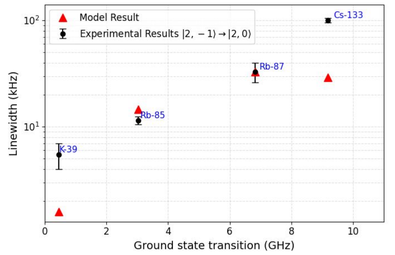
- Comparing T_2^* times as we vary dopant density and change our pumping and alignment efficiencies.
- Modeling Rubidium-85 ($I = 5/2, J = 1/2$), transition: $|2,2\rangle \rightarrow |3,3\rangle$

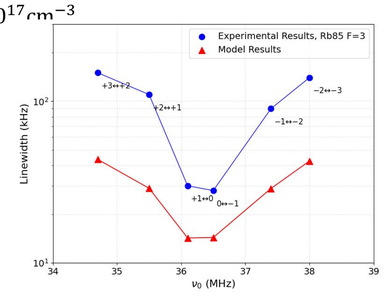




Compare to Experimental Results, High Dopant Density:

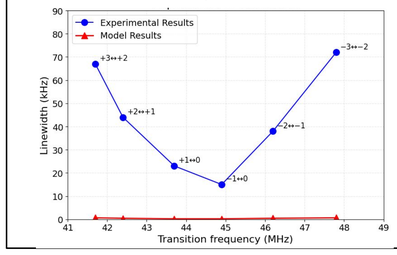
From [1]. Linewidths reported for K-39 $\rho = 5 \times 10^{16}\text{cm}^{-3}$, Rb-85/87 $\rho = 1 \times 10^{18}\text{cm}^{-3}$, and Cs-133 $\rho = 1 \times 10^{17}\text{cm}^{-3}$





Compare to Experimental Results, Low Dopant Density:

From [2]. Rb-85 F=3, $\rho = 1.5 \times 10^{16}\text{cm}^{-3}$



From [3] $\rho = 3 \times 10^{16}\text{cm}^{-3}$: Sodium-23 ($I = 3/2, J = 1/2$), transition: $|1,1\rangle \rightarrow |2,2\rangle$

- Experimental Result: 989 kHz
- Model Result: 2.29 kHz

Conclusion

My inhomogeneous broadening simulation implies that while there is a regime where spin-spin interactions between atoms ($\rho \gtrsim 10^{18}\text{atoms/cm}^3$, for Rb) becomes a dominant factor for inhomogeneous broadening, at lower densities ($\rho \sim 10^{16}\text{atoms/cm}^3$) it's subdominant. Most labs tend to grow around $\rho \sim 10^{16}\text{atoms/cm}^3$, so in this regime my model does not accurately predict inhomogeneous broadening. It is possible, however, that it provides a lower limit on broadening, that if achieved, would provide more sensitive measurements. One improvement, is that some matrices, like Ne, can have spins that interact with our atoms. Adding this would be important as both low dopant density results were in crystals that may have small spin-spin interactions.

References & Acknowledgements

[1] S. Upadhyay, U. Dargyte, V. D. Dergachev, R. P. Prater, S. A. Varganov, T. V. Tscherbul, D. Patterson, and J. D. Weinstein, Spin coherence and optical properties of alkali-metal atoms in solid parahydrogen, Phys. Rev. A **100**, 063419 (2019).
[2] U. Dargyte, D. M. Lancaster, and J. D. Weinstein, *Optical and spin-coherence properties of rubidium atoms trapped in solid neon*, Phys. Rev. A **104**, 032611 (2021).
[3] A. N. Kanagin, N. de Zordo, A. Angerer, W. Kersten, N. Lagos, J. Schmiedmayer, and E. S. Redchenko, Impurities in cryogenic solids: a new platform for hybrid quantum systems, arXiv:2508.21651 (2025).
[4] D. A. Steck, *Rubidium 85 D Line Data*, available online at http://steck.us/alkalidata (revision 2.1.6, 20 September 2013).
[5] I. S. Grant and W. R. Phillips, *Electromagnetism*, 2nd ed., Manchester Physics Series (John Wiley & Sons, Chichester, 2008).